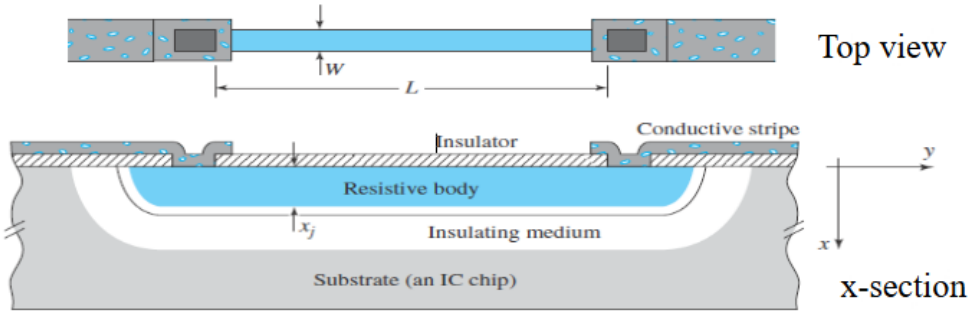
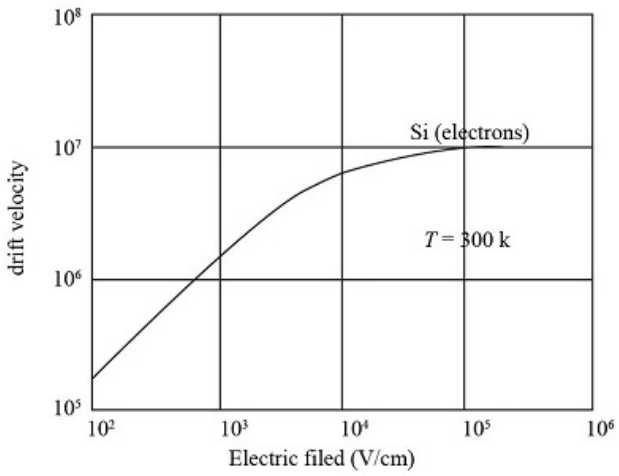
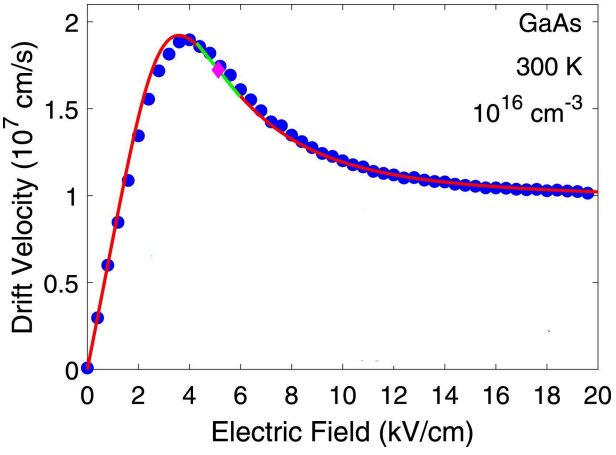
EE2021: Tutorial -2 (August 19th, 2025)

1. One way to make resistors in an IC is to use a doped region of semiconductor as follows. The dimensions are L= 1 µm, x_j=0.1 µm and W=0.5 µm. Two different resistors are made using a) Silicon and b) GaAs. The drift velocity vs electric field plot of electrons in both materials are given. Assume a uniform electron concentration of 10^{16} cm^{-3} in both cases. Plot the I-V characteristics and resistance of both resistors when voltage is varied from 0-15 V. Assume the area of cross-section of the resistor as x_j* W and length as L.







2. Using Stirling's approximation, derive the expression for the multiplicity of a large Einstein solid as

$$\Omega(N,q) \approx e^{N\ln(q/N)}e^N = \left(\frac{eq}{N}\right)^N \qquad (\text{when } q \gg N)$$

3. Prove that, for any system in equilibrium with a reservoir at temperature T, the average value of the energy is

$$-\frac{\partial}{\partial\beta}\ln Z$$

where β=1/kT

4. Imagine a particle that can be in only three states, with energies -0.05 eV, 0, and 0.05 eV. This particle is in equilibrium with a reservoir at 300 K.
   a) Calculate the partition function for this particle.
   b) Calculate the probability for this particle to be in each of the three states.
   c) Calculate the average energy of this particle.
   d) Because the zero point for measuring energies is arbitrary, we could just as well say that the energies of the three states are 0, +0.05 eV, and +0.10 eV, respectively. Repeat parts a)-c) using these numbers. Explain what changes and what doesn't.